Ibaad Muhammad

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Education

University College London Sep. 2025 – Jun. 2029
MEng Computer Science: First Class (Year 1) *London, UK*

British School Dhahran Aug. 2023 – Jun. 2025
A Levels: Mathematics A, Further Mathematics A*, Biology A, Chemistry A* *Khobar, Saudi Arabia*
Achieved highest mark in Computer Science AS Level in Eastern Province, Saudi Arabia

Experience

Histolyte Technologies - Software Engineer Intern Sep. 2026 – Present

- Contributing to software development at a digital pathology and AI healthcare startup
- Used JavaScript and CSS to diagnose and fix button, routing and formatting bugs in the company's React SPA, and refine the frontend for a more cohesive design

Eternal Gems - Freelance Full-Stack Developer Aug. 2026 – Present

- Designing and building a full-stack e-commerce platform (React, TypeScript, Node.js/Express, PostgreSQL) for a small business, implementing product catalogue, user authentication, basket, and Stripe payment integration
- Implemented secure user authentication using JWT and bcrypt password hashing, with httpOnly cookies for session storage and middleware-based route protection for authenticated pages
- Leveraging AI tools like Claude Code to brainstorm approaches and verify code

Projects

Habitly | *Python, Flask, JavaScript, React, Tailwind CSS, SQLite, SQLAlchemy, Git* Jan. 2026 – Mar. 2026

- Created a full-stack activity recommendation web app with 3 coursemates using React, Flask, and SQLite
- Crafted a heuristic scoring algorithm combining user state inputs (physicality, sociability, time available) with time of day, recency bias and live weather data from the Open-Meteo API to tailor activity recommendations
- Architected the database schema and implemented secure user authentication with hashed password storage

Rolls-Royce Reactor Modelling | *Python, NumPy, pandas, Matplotlib, pytest, Flake8* Oct. 2025 – Jan. 2026

- Selected by Rolls-Royce and elected by peers to lead a team of 5 in constructing a Python-based high temperature gas reactor simulation model, projecting ∼150 tonnes of CO_2 reduced daily
- Incorporated supervisor feedback from an interim presentation to adopt a structured, criteria-based scoring framework for evaluating design options, improving decision documentation across the project
- Validated model accuracy through automated unit testing, literature comparison, and programmatic graphing, achieving results within 1% of published values

Typocalypse | *Unity, C#, Git* Sep. 2025 – Feb. 2026

- Developed a 2D cross-platform arcade game in Unity with C#, applying object-oriented programming principles, where players advance by typing to evade AI-controlled zombies, garnering hundreds of impressions
- Implemented online multiplayer using Unity Netcode with cloud matchmaking and relay services, debugging synchronisation issues including a race condition in the client joining sequence
- Designed the in-game art, sound effects and UI/UX to create a cohesive visual and audio experience

Technical Skills

Languages: Python, C, C++, C#, Java, SQL, HTML, CSS, JavaScript, TypeScript
Frameworks/Technologies: Flask, React, Node.js, NumPy, pandas, Matplotlib, PostgreSQL, Express, pytest, SQLite
Developer Tools: Git, Docker, VS Code, PyCharm, IntelliJ IDEA, Claude Code, WSL, Unity, Vercel, Supabase

Achievements & Interests

- Published several original video games, totalling ∼1000 impressions and generating revenue
- Awarded Gold and achieved highest score in school in the UKMT Senior Mathematical Challenge
- Paid Computer Science Outreach Ambassador at UCL, teaching younger students fundamental computing concepts through engaging workshops
- Selected for Expedia Group's Career Discovery Programme (2026), following CV screening and assessment
- Solved 400+ LeetCode problems in topics including trees, dynamic programming, graphs and sliding window
- Competed in multiple hackathons and game jams, prototyping functional applications under strict time constraints